1 Even strong traces and unweighted first roots

Two new fixed ordinary polynomials have fourteen unrestricted integer unknowns:

$$(\deg, \mathrm{ops}) = (85, 110), \qquad (87, 108).$$

The charged binary-operation counts are respectively $(63, 24, 23)$ and $(62, 23, 23)$ for multiplication, addition and subtraction. They strictly improve the matched $(86, 112)$ and $(88, 110)$ points; $(82, 113)$ remains a separate tradeoff. All six full fixed numeral payloads and their finite reconstruction recipe are retained. These are ordinary reductions through the inherited native/compiler/ refutation theorem, not a consistency assertion, global optimum, or new end-to-end proof-assistant certificate.

1.1 Complete polynomial definitions

The unknowns are

$$j, C, z, h, r, v, w, T, c, n, f, y, p_2, p_3 \in \mathbb{Z}.$$

Use the authentic ports $\beta, K_0, M_C, M_F, A_c, B_c$, with M_F already shifted, $\beta = 2^{1064921} - 1$, $A_c = 2^{1113}$ and $B_c = 2^{666104}$. Define

$$\begin{gathered}
J = 6j + 1, \quad q_m = \beta J, \quad q = q_m + 1, \quad W = 12w + 4, \quad X = qW, \quad Y = v, \\
F = (K_0 + W)C - zq_m, \quad R = [q(q - F) - C](q^2 - 1) + (M_C + qM_F)J, \\
E = XY, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \quad \ell = XY^2, \\
\eta = c - Yk, \quad L = \ell k, \quad I = (8r + 4)c^2, \quad Q = \Delta(f^2 - 1), \\
V = f(cT - R) - c, \quad M = 2q^3, \quad U = MH, \quad K = X(2X - 3c) - 2, \\
Z_0 = Un + MK + H(Y - q^3), \quad D_{\mathrm{err}} = (A_cF - q)^2 + (B_cC - q)^2.
\end{gathered}$$

Choose the chart

deg	k	$\mathcal{B}$	Z
85	$R + 1 + h\ell$	$(Yk)^2(q^2 - D_{\mathrm{err}}) - q^2[4(Y\eta)^2 + (kq^2)^2]$	$Z_0 + UL$
87	$R + 1 + hE$	$(Yk)^2(q^2 - D_{\mathrm{err}}) - q^2[4(Y\eta)^2 + (kq^3)^2]$	Z_0

and finish with

$$\begin{aligned}
G &= \mathcal{B} - 1 - p_2^2 - p_3(p_3 + 1), & S &= f^2 - \Delta I^2, \\
u_u &= (V + y)^2, & E_a &= u_u - Q(u_u - V^2), \\
D_\tau &= 1 + L(L + k), & & \\
A &= U^2G, & B &= U^2D_\tau, \\
N &= (Z^2 - A - B)^2 - 4AB, & P &= (N + 1)SE_a - 1.
\end{aligned}$$

The identity $E_a = V^2 - (Q - 1)y(2V + y)$ shows that sharing u_u saves one charged multiplication without changing the polynomial. Every name denotes a polynomial abbreviation, not an additional unknown. The 87-chart evaluator uses $L = E(Yk)$ and does not separately compute ℓ.

1.2 Unweighted extraction, including all exceptional branches

Unconditionally q is even and nonzero, $|W| \geq 4$, $|X| \geq 8$, H is odd and nonzero, and $U \neq 0$. Since $a = (X + 1)Y$ is zero or has $|a| \geq 7$, $\Delta = (a + 2)^2 - 1 \geq 3$. Also ℓ is even, so $D_\tau = 1 + \ell(\ell + 1)k^2 \geq 1$.

At $P = 0$ the three integer factors are units. As N is a square modulo four, $N + 1 \neq -1$; as $4 \mid I$, $S \equiv f^2 \pmod{16}$ and $S \neq -1$. Therefore $N = 0$, $S = 1$, $E_a = 1$. With $\rho = Z/U$,

$$(\rho^2 - G - D_\tau)^2 = 4GD_\tau.$$

This forces $G \geq 0$, including $G = 0$. Hence $\mathcal{B} > 0$ and $Yk \neq 0$. Since $|\ell| \geq 8$ and $|X| \geq 4|q|$,

$$D_\tau \geq 1 + \tfrac{1}{2}X^2Y^4k^2 \geq 1 + 8q^2Y^4k^2 > q^2Y^2k^2 \geq \mathcal{B} > G.$$

Thus $\rho \neq 0$ before any division by it. The rationals

$$p = \frac{\rho^2 + G - D_\tau}{2\rho}, \qquad \tau = \rho - p$$

satisfy $p^2 = G$, $\tau^2 = D_\tau$. Each is integral because a rational with integral square is integral. This proof covers zero and coincident square classes; no generic independence hypothesis occurs. The universal signed factorization is

$$(x^2 - A - B)^2 - 4AB = \prod_{\epsilon_1, \epsilon_2 \in \{\pm 1\}} (x - \epsilon_1\sqrt{A} - \epsilon_2\sqrt{B}).$$

Write $s = 1$ for degree 85 and $s = 0$ for degree 87. Restore the even-trace fifteen-variable kernel by

$$p_1 = p, \qquad n_{\mathrm{old}} = n + sL - (2L + k) - p + \tau.$$

Its carrier is

$$\mathcal{A} = (Z_{\mathrm{old}} + Uk)^2 + 4U(LZ_{\mathrm{old}} - U) = (Z_{\mathrm{old}} + U(2L + k))^2 - 4U^2 D_\tau,$$

and therefore vanishes. The remaining rows are $S - 1 = 0$, $E_a - 1 = 0$ and $\mathcal{B} - 1 - p_1^2 - p_2^2 - p_3(p_3 + 1) = 0$. Conversely any zero of this kernel has an integral τ with $Z_{\mathrm{old}}/U + 2L + k = 2\tau$: the rational on the left has square $4D_\tau$, so is an even integer. Delete p_1 and set

$$n = n_{\mathrm{old}} - sL + (2L + k) + p_1 - \tau.$$

Then $\rho = p_1 + \tau$ and $P = 0$. These are total maps on zeros, not claimed polynomial automorphisms; the root-dependent integral shift removes one gate.

1.3 Signed unsquared trace: the new parity argument

The comparison signed kernel has odd strong scale $(2r_{\mathrm{old}} + 1)c^2$ and squared trace $f_{\mathrm{old}}^2(cT_{\mathrm{old}} - R) - c$. Its retained proof is `inherited_round20/inherited/PROOF.md` for the q^6/XY^2 chart, and `FREE_SCALE15_PROOF.md` in its inherited roots source for q^8/XY. The normalized rank and auxiliary lemmas are in the same source's `06_PROOF.md`. We verify the changed hypotheses before restoring a literal old zero.

The unchanged carrier and guard give $c \neq 0$, the strict centered strip, $q_0 = |q| \geq 16$, $x = |X| \geq 4q_0$ and

$$q_0(q_0^2 - 1) < R < q_0^4 - 2.$$

They give $|Y| > q_0^2$, $|XY^2| > 3R + 1$ in the 85 chart, and $|Y| > q_0^3$, $|XY| > 3R + 1$ in the 87 chart. Carrier integrality gives even Y and the exact divisibilities $M \mid H(Y - q^3)$, $H \mid MK$ without premature cancellation. Put $B_0 = |a + 2|$, $c_0 = |c|$, $I_+ = |8r + 4|c_0^2$ and $f_+ = |f|$. Then $I_+ > 0$ is even, $B_0 \geq 3$, and

$$f_+^2 - (B_0^2 - 1)I_+^2 = 1, \qquad f_+ > 2I_+.$$

Pell classification gives $f_+ = \chi_{B_0}(m)$, $I_+ = \psi_{B_0}(m)$. The residue $V \equiv -c \pmod{f_+}$ is unchanged. The ordinary odd auxiliary index and folding lemma therefore recover $c_0 = \psi_{B_0}(p)$ before any residue modulo c_0 is needed: fold the auxiliary index to $0 \leq p \leq m$, and use $\psi_{B_0}(p) + c_0 \leq 2I_+ < f_+$ to turn its signed congruence into equality. The first Pell classification and centered bounds give $p > n_0 \geq 24$ and $c_0 > 2p, 2R$.

The normalized rank lemma gives $m = p\nu$ and $c_0 \mid \nu$. Indeed, strong divisibility gives $p \mid m$; with $D = \chi_{B_0}(p)$, composition gives $I_+ = c_0\psi_D(\nu)$, and $\psi_D(\nu) \equiv \nu D^{\nu-1} \pmod{c_0}$ with $\gcd(D, c_0) = 1$ gives $c_0 \mid \nu$. Now $\psi_{B_0}(m) \equiv m \pmod 2$, so m is even. If c_0 is even, ν is even; if c_0 is odd, p is odd and $m = p\nu$ again makes ν even. As $D^2 \equiv 1 \pmod{c_0}$ and $\chi_D(\nu) \equiv D^\nu \pmod{c_0}$,

$$|f| = \chi_D(\nu) \equiv 1 \pmod{c_0}.$$

Consequently the unsquared trace has exactly the required signed residues

$$V \equiv -c \pmod{|f|}, \qquad V \equiv \pm R \pmod{c_0}.$$

The signed auxiliary step-down gives an odd index λ with $\lambda \equiv \pm p \pmod m$ and $\lambda \equiv \pm R \pmod{c_0}$. Since $c_0 \mid m$ and $0 < p, R < c_0/2$, it follows that $p = R$. Thus the extra sign does not invalidate the argument. The unchanged surviving first-index branch gives $2n_0 = R + 1$. Whole-product projection, the modulo-three orientation and late integer cancellation then give

$$c, k, Y > 0, \quad X = 2^R, \quad R \text{ odd}, \quad Y = (2v_{\mathrm{scale}} + 1)q^3, \quad H \mid K, \quad 0 < \eta/k < 1/4.$$

These steps use only the already established indices, carrier and strip, not the old square on f. In particular $B_0 = a + 2 \equiv 2 \pmod 8$ and c is odd.

1.4 Fresh strong witnesses: soundness and completeness

One cannot keep f, T when returning to an odd-scale source. Let

$$m_0 = Rc, \qquad f_0 = \chi_{B_0}(m_0), \qquad I_0 = \psi_{B_0}(m_0).$$

Composition gives $c^2 \mid I_0$; as R, c, m_0 are odd, I_0/c^2 is odd. Thus $r_{\mathrm{old}} = (I_0/c^2 - 1)/2$ restores the old strong row. For every odd $R = 2d + 1$, the integer quotient polynomials obey

$$\chi_Z(2d + 1) = ZQ_d(Z^2), \quad Q_d(0) = (-1)^d(2d + 1), \quad Q_d(1 - B_0^2) = (-1)^d\psi_{B_0}(2d + 1).$$

They follow from $Q_0 = 1$, $Q_1 = 4x - 3$ and $Q_{d+1} = (4x - 2)Q_d - Q_{d-1}$. Set $S_0 = \Delta I_0$ and

$$V_0 = (-1)^{d+1}Q_d(S_0^2), \qquad u_0 = \psi_{S_0}(R).$$

Then $S_0^2V_0^2-(S_0^2-1)u_0^2=1$, while $V_0\equiv -c \pmod{f_0^2}$ and $V_0\equiv -R \pmod{c}$. Since $\gcd(c,f_0)=1$ and $f_0^2\equiv 1 \pmod{c}$,

$$T_{\text{old}}=\frac{V_0+c+Rf_0^2}{cf_0^2},\qquad y_{\text{old}}=u_0-V_0$$

are integers and restore the squared auxiliary trace. Every outer, carrier and guard coordinate is unchanged. This is a literal zero of the correct old signed kernel (separately for each chart), not an assumed $R\equiv 3 \pmod 4$ branch. Its authenticated native/compiler theorem applies.

Conversely every old zero has the same established outer data. Double the seed:

$$f=2f_0^2-1,\qquad I=2f_0I_0.$$

Because $B_0\equiv 2 \pmod 4$ and m_0 is odd, $f_0\equiv 2 \pmod 4$. Hence $I/(4c^2)=(f_0/2)(I_0/c^2)$ is odd, so $r=(I/(4c^2)-1)/2$ is integral and $I=(8r+4)c^2$. The doubled pair has norm one; $f_0^2\equiv 1 \pmod c$ directly gives $f\equiv 1 \pmod c$. Using the same signed quotient construction at $S=\Delta I$ yields V_*,u_*; now

$$T=\frac{V_*+c+Rf}{cf},\qquad y=u_*-V_*$$

are integers because the numerator is divisible by the coprime integers c,f. This gives $V=f(cT-R)-c=V_*$ and the new auxiliary row. No positivity of T is required on the unrestricted-integer domain. Apply the unweighted root map to finish. Both directions are defined on all zeros using fresh witnesses.

1.5 Attained degrees and exact coefficients

Put

$$q_0=6\beta j,\ X_0=12q_0w,\ Y_0=v,\ a_0=X_0Y_0,\ F_0=12wC-q_0z,\ R_0=-q_0^3F_0,\ \ell_0=X_0Y_0^2,\ U_0=8q_0^3a_0.$$

Both complete strong/auxiliary tops are

$$[S]_{12}=-64a_0^2r^2c^4,\qquad [E_a]_{15}=2a_0^2f^3R_0y.$$

For degree 87,

$$[N]_{60}=U_0^4R_0^4(q_0^8+\ell_0^2)^2,\qquad [j^{56}w^8v^8z^5r^2c^4f^3y]P_{87}=-2^{91}3^{64}\beta^{56}\neq 0.$$

For degree 85 retain $k_0=R_0+h\ell_0$ and define

$$T_0=A_c^2Y_0^2F_0^2+4q_0^2Y_0^4+q_0^6.$$

The entire cancellation

$$(Z_0+UL)^2-U^2D_\tau=Z_0^2+2ULZ_0-U^2(1+Lk)$$

has degree 26. Thus the square in N has degree 56 and the other term has top

$$[N]_{58}=4U_0^4\ell_0^2k_0^4T_0,\qquad [j^{48}w^{10}v^{12}z^5r^2c^4f^3y]P_{85}=-2^{89}3^{58}\beta^{48}\neq 0.$$

These full leading forms, with respectively 18 and 70 monomials after expansion, prove the exact degrees on the supplied fixed numeral slice, not merely generic or sampled degrees.

1.6 A separate correlated-root separation lemma

Let $F,D\equiv 1 \pmod 3$, $G\in\mathbb{Z}$, $3\mid Q$ and $b,c\in\mathbb{Z}$. For algebraic roots

$$f^2=F,\quad \tau^2=D,\quad t^2=\frac{Q(fb-c)^2-1}{Q-1},\quad g^2=G,$$

a rational $\rho=f+3\tau+9t+27g$ forces all four roots rational. Work in the integral closure of $\mathbb{Z}_{(3)}$ in a normal field containing all roots. They are integral over this ring, since $Q-1$ is a unit and $t^2=\alpha+\beta f$ with $\alpha,\beta\in\mathbb{Z}_{(3)}$. If an automorphism flips f, subtract the trace equations to obtain $f=3u$ with integral u. But $u^2=F/9$ is rational and integral over $\mathbb{Z}_{(3)}$, hence lies in that ring, contradicting $F\equiv 1 \pmod 3$. Thus all automorphisms fix f, so f is rational. Divide the conjugate difference by three to repeat for τ; then t^2 is rational and a three-adic unit, so divide by nine and repeat for t. Finally g is rational. This includes imaginary roots and dependent square classes. A valid norm must use the different auxiliary roots associated to each sign of f; the independent four-root norm cannot simply be reused. The explored twelve-variable evaluators remained dominated, so no new twelve-variable frontier claim is made.

1.7 Verification and dependency boundary

The standard-library checker binds every literal register independently, all inputs and six ports, topology, coverage, liveness, exact operation counts, four-factor identity, root maps, doubling, full leading cancellations and attained coefficients. It rejects individual mutations of all 110 and 108 live gates. Normal and optimized Python runs pass the same explicit conditions. The finite reconstruction recomputes all 615883 coefficient bytes, 943 tiles and 69230 basis convolution checks. All 212 inherited source files are pinned. Recurrence samples are diagnostics, not proofs of the unbounded lemmas above. The current comparison snapshot is `5274bc751a1566879ddc028a2d2a2af3c031654a`; original source attributions and licenses are preserved. The ordinary native/compiler/refutation theorem remains the named inherited dependency, not a claim inferred from finite checks.